CO1 - AT2

CSA3714 - Software Testing

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7. Payroll System:

Requirement clarity:- Unclear requirements may cause incorrect salary calculations, tax deductions, and bonus payments. The team should gather clear requirements and confirm them with users.

10. ATM Functional Defect:

The defect is a functional defect because the ATM fails to process withdrawals correctly. Test cases should check balance verification, withdrawal limits and transactional updates. Proper functional testing prevent failures.

13. Regression testing:

Historical data help identify the testing cases. The team can reuse old test case after every release to check whether existing features still work correctly. This improves test coverage and saves time

17. Testing vs Debugging:

Testing identifies defects while debugging finds and fixes their causes. For example, testing identifies an incorrect bank balance, and debugging finds the error in the transactional logic. So Debugging is always the best in terms of software quality.

19. Logistics Application - Validation

The team should test valid, invalid and missing boundaries so that logistics Validation. For example: it should reject the negative package weights. Proper validation prevent incorrect outputs and improves system reliability.

21. Healthcare System - Data consistency.

The team should perform integration testing to verify data exchange between the modules. It should compare patient records across modules and identify mismatches. Proper testing ensure accurate and consistent data.

Early Testers Involvement:

Testers should participate in requirement analysis, design, coding and testing. Early involvement helps identify defects before implementation. This reduces correction costs, delays and improves software quality.

24. Defect Repository:

A defect Repository stores defect details, causes, severity and solutions. Testers can use historical data to design better test cases and avoid repeated defects. Proper documentation improves tracking and analysis.

26. Boundary Value Testing:

Testers should check minimum, maximum and values just above and below the limits. For example, if the allowed age is 18-60, test 17, 18, 60 and 61. This helps detect boundary-related defects.

29. Defect Leakage:

Defect leakage occurs when defects escape testing and reach production. The team should improve test coverage, perform regression testing, and improve involve testers early. Regular reviews and root-cause analysis help prevent future defects.